

Chapter 4

Characterizing Analytic Functions

4.1 Power series

Theorem 4.1 (Power series expansion, Cauchy 1831). Any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\forall c \in \mathcal{D}, \forall z \in \mathcal{U}_R(c) \subseteq \mathcal{D}, \quad f(z) = \sum_{n=0}^{\infty} a_n(z-c)^n, \quad (4.1)$$

where $\mathcal{U}_R(c)$ is the open disk with its radius as R and its center at c and the coefficients are uniquely determined as

$$a_n := \frac{f^{(n)}(c)}{n!} = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} d\zeta, \quad (4.2)$$

with ρ being an arbitrary real positive number in $(0, R)$.

Proof. $|z-c| < |\zeta-c|$ holds for all $z \in \mathcal{U}_\rho(c)$. Then the geometric series yields

$$\frac{1}{\zeta-z} = \frac{1}{(\zeta-c)(1-\frac{z-c}{\zeta-c})} = \sum_{n=0}^{\infty} \frac{(z-c)^n}{(\zeta-c)^{n+1}}.$$

It follows that

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta \\ &= \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} f(\zeta) \sum_{n=0}^{\infty} \frac{(z-c)^n}{(\zeta-c)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} (z-c)^n \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} a_n(z-c)^n, \end{aligned}$$

where the first step follows from the Cauchy integral formula (Theorem 3.75), the third step from Theorems 1.77, 1.75, and D.178, and the last step from (4.2) and Theorem 3.85. Finally, the uniqueness of a_n follows from the uniqueness of the above derivation. $\square$

Exercise 4.2 (Gutzmer's inequality). With the assumptions of Theorem 4.1, show that for any $r \in (0, R)$,

$$\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq M(r)^2, \quad (4.3)$$

where $M(r)$ is defined in (3.49).

Exercise 4.3 (Alternative proof of Corollary 3.86). Prove (3.48) by using (4.3). Furthermore, show that the equality in (3.48) holds for some $n_0 \in \mathbb{N}$ if and only if $f(z) = a_{n_0}(z-c)^{n_0}$.

Example 4.4. Theorem 4.1 applied to $\text{Log}(z)$ on $\mathbb{C}_-$ in (2.5) yields a power series expansion,

$$\text{Log}(z) = \text{Log}(c) + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{c^n n} (z-c)^n, \quad (4.4)$$

where we have used the fact $(\text{Log} z)' = \frac{1}{z}$ (why?). By Theorem 1.79, the convergence radius is $|c|$. Interestingly, this example shows that the convergence radius may be bigger than the distance of the series center to the boundary of the domain of definition.

One needs to be careful about the convergence of a Taylor series. If $c \in \mathbb{C}_-$ satisfies $\text{Re } c < 0$, then $\mathbb{C}_- \cap \mathcal{U}_r(c)$ might consist of two disjoint separable connected components, where the power series converge to different functions,

$$f(z) = \begin{cases} \text{Log}(z) & \text{if } \text{Im } c > 0; \\ \text{Log}(z) + 2\pi i & \text{if } \text{Im } c < 0. \end{cases}$$

Example 4.5. $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges for $|z| = 1$, $\sum_{n=1}^{\infty} z^n$ diverges for $|z| = 1$, and the series

$$p(z) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n} \quad (4.5)$$

diverges for $z = -1$ and converges for all other z satisfying $|z| \leq 1$. Why? The answers lie in Theorem 3.110 and the fact that $p(z) = \text{Log}(1+z)$.

Example 4.6. Consider the geometric series

$$g(z) = \frac{1}{1-z^2} = \sum_{n=0}^{\infty} z^{2n}.$$

The divergence of $\sum_{n=0}^{\infty} z^{2n}$ for $|z| \geq 1, z \in \mathbb{R}$ is easily understood: $g(z)$ has two singularities at $z = \pm 1$. However the divergence of the geometric series

$$f(z) = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}.$$

for $|z| \geq 1, z \in \mathbb{R}$ is perplexing. After all, $f(z)$ is always bounded for any given $z \in \mathbb{R}$. This puzzle is solved in the context of complex analysis by observing that $f(z)$ has two singularities at $z = \pm i$. Indeed, $f(z) = g(iz)$ and the two functions are really the same complex function with different coordinates.

Exercise 4.7 (Double series theorem of Weierstrass). Let the power series

$$f_j(z) = \sum_{k=0}^{\infty} c_{jk}(z-c)^k \quad (4.6)$$

be convergent in the disk $\mathcal{U}_r(c)$ with $r > 0$. If the series $F := \sum_{j=0}^{\infty} f_j$ converges normally in $\mathcal{U}_r(c)$, show that F is analytic in $\mathcal{U}_r(c)$ and can be represented as

$$F(z) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{\infty} c_{jk} \right) (z-c)^k. \quad (4.7)$$

Example 4.8. The following $\mathcal{C}^\infty$ function due to Cauchy

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2}), & x \neq 0; \\ 0, & x = 0 \end{cases} \quad (4.8)$$

satisfies $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$, but it does not have a Taylor expansion at any neighborhood at 0.

Theorem 4.9 (Equivalence of analytic functions on open sets). The following assertions are equivalent for a function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open.

- (A) f is analytic at any point $z \in \mathcal{D}$.
- (B) f is totally differentiable in the sense of real analysis ($\mathbb{C} = \mathbb{R}^2$) and satisfies the Cauchy–Riemann equations (2.32).
- (C) f is continuous and for each triangle path $\langle z_1, z_2, z_3 \rangle$ whose convex hull is contained in $\mathcal{D}$ we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0.$$

- (D) f locally admits primitives, i.e., for each point $c \in \mathcal{D}$ there exists an open disk $\mathcal{U}(c) \subset \mathcal{D}$ such that $f|_{\mathcal{U}(c)}$ has a primitive.
- (E) f is continuous, and for each disk $\mathcal{U}_\rho(c)$ satisfying $\overline{\mathcal{U}_\rho(c)} \subset \mathcal{D}$ we have

$$\forall z \in \mathcal{U}_\rho(c), \quad f(z) = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta.$$

- (F) f is locally representable as a convergent power series.
- (G) f is representable as a power series in each open disk fully contained in $\mathcal{D}$.

Proof. We list the main points of the proof as follows.

- $(A) \Leftrightarrow (B)$: Cauchy–Riemann theorem 2.58;
- $(A) \Rightarrow (C)$: Cauchy theorem 3.55;
- $(C) \Rightarrow (A)$: Morera’s theorem 3.93;
- $(C) \Rightarrow (D)$: Theorem 3.42 and Lemma 3.58;
- $(D) \Rightarrow (C)$: Theorem 3.42;
- $(A) \Rightarrow (E)$: Cauchy integral formula 3.75;
- $(E) \Rightarrow (F)$: Taylor theorem 4.1;

- $(F) \Rightarrow (A)$: Corollary 3.106;
- $(A) \Rightarrow (G)$: Taylor theorem 4.1;
- $(G) \Rightarrow (F)$: A universal statement implies a existential statement. $\square$

Exercise 4.10. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_r(0)$. Show that if f is nonconstant and $\sum_{n=2}^{\infty} n|a_n|r^{n-1} \leq |a_1|$, then f is injective.

Exercise 4.11. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_R(0)$. Define $A(r) = \max_{|z|=r} \operatorname{Re} f(z)$, Show that for each $r \in (0, R)$ and $n \in \mathbb{N}^+$,

- (a) $a_n r^n = \frac{1}{\pi} \int_0^{2\pi} [\operatorname{Re} f(re^{i\theta})] e^{-in\theta} d\theta$;
- (b) $|a_n| r^n \leq 2A(r) - 2\operatorname{Re} f(0)$.

4.2 Elementary domains

Definition 4.12. A domain $\mathcal{D} \subset \mathbb{C}$ is called an *elementary domain* iff any analytic function defined on $\mathcal{D}$ has a global primitive in $\mathcal{D}$.

Lemma 4.13. Let $\mathcal{D}$ and Ω be two elementary domains. If $\mathcal{D} \cap \Omega$ is non-empty and connected, then $\mathcal{D} \cup \Omega$ is also an elementary domain.

Proof. Let $f : \mathcal{D} \cup \Omega \rightarrow \mathbb{C}$ be analytic. We need to show that f has a primitive. Since $\mathcal{D}$ and Ω are both elementary domains, f has a primitive $F_1 : \mathcal{D} \rightarrow \mathbb{C}$ on $\mathcal{D}$ and a primitive $F_2 : \Omega \rightarrow \mathbb{C}$ on Ω . The difference $F_1 - F_2$ must be locally constant in $\mathcal{D} \cap \Omega$, and therefore constant since $\mathcal{D} \cap \Omega$ is connected. By addition of a constant if necessary, we can assume WLOG that $F_1|_{\mathcal{D} \cap \Omega} = F_2|_{\mathcal{D} \cap \Omega}$. Hence the functions F_1 and F_2 now glue to a single function $F : \mathcal{D} \cup \Omega \rightarrow \mathbb{C}$. $\square$

Corollary 4.14. Slit annuli are elementary domains.

Proof. By Example 3.63, the ring sector is a star-shaped domain in the sense of Definition 3.62 and thus an elementary domain. By Lemma 4.13, the union of two ring sectors are also an elementary domain if their intersection is non-empty and connected. Hence a ring sector can be extended to a slit annulus while retaining the property of elementary domains. $\square$

Corollary 4.15. Let $\mathcal{D}_1 \subset \mathcal{D}_2 \subset \cdots$ be an increasing sequence of elementary domains. Then their union $\cup_{n=1}^{\infty} \mathcal{D}_n$ is also an elementary domain.

Proof. This follows directly from Lemma 4.13. $\square$

Lemma 4.16. Suppose $\mathcal{D} \subset \mathbb{C}$ is an elementary domain and $\varphi : \mathcal{D} \rightarrow \mathcal{D}^*$ a globally conformal map in the sense of Definition 2.39. Then $\mathcal{D}^*$ is also an elementary domain.

Proof. It suffices to show that any analytic function $f_* : \mathcal{D}^* \rightarrow \mathbb{C}$ has a primitive F_* satisfying $F'_* = f_*$. The condition of φ being globally conformal, Definition 2.39, Theorem 2.40, Theorem 2.25, and Theorem 3.85 imply that φ , φ^{-1} , and φ' are all analytic. Hence $(f_* \circ \varphi)\varphi' : \mathcal{D} \rightarrow \mathbb{C}$ is analytic. Since $\mathcal{D}$ is an elementary domain, there exists

$F : \mathcal{D} \rightarrow \mathbb{C}$ such that $F' = (f_* \circ \varphi)\varphi'$. Hence we can construct $F_* : \mathcal{D}^* \rightarrow \mathbb{C}$ by $F_* = F \circ \varphi^{-1}$ such that

$$(F \circ \varphi^{-1})' = (F' \circ \varphi^{-1})\frac{1}{\varphi'} = f_*. \quad \square$$

Exercise 4.17. Show that the sickle-shaped domain $\mathcal{D} = \mathcal{U}_1(0) \setminus \overline{\mathcal{U}_{\frac{1}{2}}(\frac{1}{2})} = \{z \in \mathbb{C} : |z| < 1, |z - \frac{1}{2}| > \frac{1}{2}\}$ is an elementary domain.

Hint: Consider the conformal map $z \mapsto \frac{1}{1-z}$. What is the image of $\mathcal{D}$ under this map?

4.3 Zeros of analytic functions

Definition 4.18. A point $a \in \Omega$ is called a *zero* or a *root* of an analytic function $f : \Omega \rightarrow \mathbb{C}$ iff $f(a) = 0$. A zero point a of f is of *multiplicity* m iff $\exists m \in \mathbb{N}^+$ such that

$$f^{(m)}(a) \neq 0 \text{ and } \forall j = 0, 1, \dots, m-1, f^{(j)}(a) = 0.$$

In particular, a zero of multiplicity 1 is a *simple zero* or a *simple root*.

Example 4.19. The zeros of $f(z) = \sin z - 1$ form the set $\{\frac{\pi}{2} + 2k\pi : k \in \mathbb{Z}\}$. Each zero is of multiplicity 2 and thus a critical point of f .

Lemma 4.20. Let Ω be a domain and $a \in \Omega$ a zero point of an analytic function $f : \Omega \rightarrow \mathbb{C}$. Then the multiplicity of f at a is m if and only if $f(z) = (z-a)^m \varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$.

Proof. This follows directly from Theorem 4.1. $\square$

Exercise 4.21. Let f be an analytic function in $\mathcal{U}_R(0)$ with $R > 0$. Show that if there exist constants $\rho \in (0, 1)$ and $C > 0$ such that for sufficiently large $n \in \mathbb{N}$, it holds $|f(\frac{1}{n})| \leq C\rho^n$, then $f \equiv 0$.

4.3.1 Roots of a nonzero analytic function are discrete

Definition 4.22. A subset $\mathcal{M} \subseteq \mathcal{X}$ in a metric space $(\mathcal{X}, d)$ is *discrete* iff

$$\forall x \in \mathcal{X}, \exists \epsilon > 0 \text{ s.t. } \forall y \in \mathcal{M} \setminus \{x\}, d(x, y) > \epsilon. \quad (4.9)$$

Such a set consists of *isolated points*.

Lemma 4.23. A subset $\mathcal{M} \subseteq \mathcal{X}$ is *discrete* in $\mathcal{X}$ if and only if there is no limit point of $\mathcal{M}$ in $\mathcal{X}$.

Proof. By Definition 2.16, the existence of a limit point x of $\mathcal{M}$ in $\mathcal{X}$ implies

$$\exists x \in \mathcal{X} \text{ s.t. } \forall \epsilon > 0, \exists y \in \mathcal{M} \text{ s.t. } d(x, y) \in (0, \epsilon).$$

The negation of the above statement yields (4.9). $\square$

Example 4.24. The set $\mathcal{M} := \{\frac{1}{n} : n \in \mathbb{N}^+\}$ is not discrete in $\Omega = \mathbb{C}$, but is indeed discrete in $\Omega = \mathbb{C} \setminus \{0\}$.

Example 4.25. The zeros of $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0, \end{cases} \quad (4.10)$$

are $0, \pm \frac{1}{n\pi}$ where $n \in \mathbb{N}^+$ and 0 is a limit point of the zeros.

Example 4.26. Combine (4.8) and (4.10) and we have a C^∞ function whose zeros have a limit point in $\mathbb{R}$,

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2}) \sin \frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0. \end{cases} \quad (4.11)$$

Theorem 4.27. The set of zeros of a nonzero analytic function $f : \Omega \rightarrow \mathbb{C}$ with $\Omega \subset \mathbb{C}$ open is discrete in Ω .

Proof. Let a be a zero of f with multiplicity m . Lemma 4.20 yields $f(z) = (z-a)^m \varphi(z)$ where $\varphi(a) \neq 0$ and φ is analytic in $\mathcal{U}_r(a)$ for some $r > 0$. The continuity of φ gives

$$\exists R > 0 \text{ s.t. } \forall z \in \mathcal{U}_R(a), \varphi(z) \neq 0.$$

Then the proof is completed by Definition 4.22. $\square$

4.3.2 Local mapping behaviors

Theorem 4.28. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function on an elementary domain, and $f(z) \neq 0$ for all $z \in \mathcal{D}$. There exists an analytic function $h : \mathcal{D} \rightarrow \mathbb{C}$ such that

$$f(z) = \exp(h(z)), \quad (4.12)$$

where the function h is called an *analytic branch of the logarithm* of f .

Proof. By Theorem 3.85, f being analytic implies that f' is also analytic. Hence $\frac{f'}{f}$ is also analytic and by Definition 4.12 it has a primitive, say F . Then the function $G(z) := \frac{\exp(F(z))}{f(z)}$ satisfies

$$\forall z \in \mathcal{D}, G'(z) = \frac{\frac{f'}{f} \exp(F)f - \exp(F)f'}{f^2} = 0.$$

Hence for all $z \in \mathcal{D}$ we have $\exp(F(z)) = Cf(z)$ with $C \neq 0$. Set $C = \exp(c)$ and the following function satisfies (4.12):

$$h(z) = F(z) - c. \quad \square$$

Corollary 4.29. Under the assumption of Theorem 4.28, there exists for any $n \in \mathbb{N}$ an analytic function $H : \mathcal{D} \rightarrow \mathbb{C}$ such that $H^n = f$.

Proof. Set $H(z) = \exp(\frac{1}{n}h(z))$ and use Theorem 4.28. $\square$

Theorem 4.30 (Local behavior of an analytic function at a root). For a nonconstant analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ where $\mathcal{D}$ is a domain, $0 \in \mathcal{D}$, and $f(0) = 0$, there exists $r > 0$ and a globally conformal map $g : \mathcal{U}_r(0) \rightarrow \mathbb{C}$ such that

$$\forall z \in \mathcal{U}_r(0), f(z) = [g(z)]^m \quad (4.13)$$

where m is the multiplicity of the root of f at 0.

Proof. Set $a = 0$ in Lemma 4.20 and we have

$$\exists R > 0 \text{ s.t. } \forall z \in \mathcal{U}_R(0), \quad f(z) = z^m \varphi(z), \quad (4.14)$$

where φ is analytic in $\mathcal{U}_R(0)$ and $\varphi(0) \neq 0$. The continuity of φ implies the existence of $r \in (0, R]$ such that $\varphi(z) \neq 0$ for all $z \in \mathcal{U}_r(0)$. By Corollary 4.29, there exists an analytic function h such that $h^m = \varphi$. Then (4.13) follows from

$$g(z) := zh(z).$$

Clearly, $g : \mathcal{U}_r(0) \rightarrow \mathbb{C}$ is analytic. For g to be globally conformal, it suffices to show $g'(0) \neq 0$, cf. Theorems 2.40 and 4.27. Theorem 4.1 gives $\varphi(z) = \sum_{j=0}^{\infty} a_{m+j} z^j$ and thus

$$f(z) = z^m \sum_{j=0}^{\infty} a_{m+j} z^j, \quad (4.15)$$

where $a_m \neq 0$ follows from Definition 4.18. On the one hand, (4.15) yields $f^{(m)}(0) = m!a_m$. On the other hand, (4.13) gives $f^{(m)}(0) = m!(g'(0))^m$. Then $0 \neq a_m = (g'(0))^m$ implies $g'(0) \neq 0$. $\square$

4.3.3 The identity theorem

Theorem 4.31 (Identity). For two analytic functions $f, g : \mathcal{D} \rightarrow \mathbb{C}$ defined on a nonempty domain $\mathcal{D}$, the following statements are equivalent:

- (a) $f = g$.
- (b) The coincidence set $\{z \in \mathcal{D} : f(z) = g(z)\}$ has a limit point in $\mathcal{D}$.
- (c) $\exists a \in \mathcal{D}$ s.t. $\forall n \in \mathbb{N}, f^{(n)}(a) = g^{(n)}(a)$.

Proof. The equivalence of (a) and (b) follows from applying Theorem 4.27 to $f - g$ and that of (a) and (c) from applying Theorems 4.27 and 4.1. $\square$

Lemma 4.32. If the product of two analytic functions on a domain $\mathcal{D}$ vanishes identically, then one of the two functions vanishes identically on $\mathcal{D}$.

Proof. Suppose $f(z)g(z) = 0$ for all $z \in \mathcal{D}$. We show $f \neq 0$ implies $g = 0$. $f \neq 0$ gives $\exists a \in \mathcal{D}$ s.t. $f(a) \neq 0$. Then the continuity of f yields

$$\exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(a), \quad f(z) \neq 0.$$

Hence $fg = 0$ and Theorem 4.27 imply $g(z) = 0$ for all $z \in \mathcal{U}_r(a)$. The proof is completed by Theorem 4.31. $\square$

Exercise 4.33. Decide, whether there are analytic functions $f, g : \mathcal{U}_R(0) \rightarrow \mathbb{C}$, with

- (a) $f(\frac{1}{n}) = f(-\frac{1}{n}) = \frac{1}{n^2}, n \geq 1$;
- (b) $g(\frac{1}{2n-1}) = g(\frac{1}{2n}) = \frac{1}{2^n}, n \geq 1$.

Exercise 4.34. Let f be an analytic function on a domain $\mathcal{D}$ satisfying that for any $z \in \mathcal{D}$, there exists an integer $n = n(z) \geq 0$ such that $f^{(n)}(z) = 0$. Show that f is a polynomial.

Hint: $\mathcal{D}$ can be represented as $\cup_{n=0}^{\infty} g_n^{-1}(\{0\})$, where $g_n(z) = f^{(n)}(z)$.

Notation 3. $H(\mathcal{D})$ denotes the set of analytic functions on a nonempty open subset $\mathcal{D} \subset \mathbb{C}$. $\mathcal{O}(\mathcal{D})$ denotes the set of analytic functions on a domain $\mathcal{D} \subset \mathbb{C}$.

Lemma 4.35. $H(\mathcal{D})$ is a commutative ring with unity.

Proof. By Definition G.32, $H(\mathcal{D})$ is clearly a ring with unity since the constant function $\mathbf{1}$ is the multiplication identity. The commutativity of the functions follows from that of complex numbers. $\square$

Exercise 4.36. If $\mathcal{D}$ is not connected, then $H(\mathcal{D})$ might have divisors of zero. Give one such example.

Theorem 4.37. $H(\mathcal{D})$ is an integral domain if and only if $\mathcal{D}$ is a domain.

Proof. For a nonempty open set $\mathcal{D} \subseteq \mathbb{C}$, Lemma 4.35 states that $H(\mathcal{D})$ is a commutative ring with unity, where the unity $\mathbf{1}$ is different from the additive identity $\mathbf{0}$.

The sufficiency follows from Definition G.40 and the fact of $H(\mathcal{D})$ having no divisors of zero, thanks to Lemma 4.32 and Definition G.34.

As for the necessity, the only difference between an integral domain and a commutative ring with unity $\mathbf{1} \neq \mathbf{0}$ is that the integral domain contains no divisors of zero. Suppose $\mathcal{D}$ in Notation 3 is not connected. Then Exercise 4.36 leads to a contradiction, which completes the proof. $\square$

Example 4.38. The unit disk $\mathbb{D}$ is a domain, but the analytic function $f(z) = z$ is zero at $z = 0$ and thus does not have a multiplicative inverse in $\mathcal{O}(\mathbb{D})$.

4.3.4 The open mapping theorem

Definition 4.39. A function is *open* iff it maps open sets to open sets.

Example 4.40. The function $f(x) = \sin x$ is $\mathcal{C}^\infty$ and $\mathbb{R}$ is open, but $\sin(\mathbb{R}) = [-1, 1]$ is not open.

Theorem 4.41 (Open mapping). Any nonconstant analytic function on a domain is open.

Proof. The function $p(z) = z^n$ maps an open disk of radius r onto to an open disk of radius r^n : $re^{i\varphi} \mapsto r^n e^{in\varphi}$. At each point z of an open set Ω , there exists an open disk $\mathcal{U}_r(z)$ at z such that $\mathcal{U}_r(z) \subset \Omega$. Then Ω can be considered as the union of open disks and Definition 4.39 implies that p is open. The rest of the proof follows from Theorem 4.30. and the fact that the function g is a globally conformal map and thus a homeomorphism. $\square$

Corollary 4.42. Let $\mathcal{D}$ be a domain and $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function. If $\operatorname{Re} f$ or $\operatorname{Im} f$ or $|f|$ is a constant function, then f is a constant function.

Proof. If $\operatorname{Re} f$ or $\operatorname{Im} f$ or $|f|$ is a constant, $f(\mathcal{D})$ cannot be an open set in $\mathbb{C}$, and thus Definition 4.39 and $\mathcal{D}$ being open imply that f is not an open map. The rest of the proof follows directly from the contraposition of Theorem 4.41. $\square$

Exercise 4.43. Show that if $f : \mathcal{D} \rightarrow \mathcal{D}$ is analytic on a domain $\mathcal{D}$ and satisfies $f \circ f = f$, then f is either constant or the identity map.

where the last step follows from $\bar{h} \rightarrow 0 \Leftrightarrow h \rightarrow 0$ for $h \in \mathbb{C}$ and the third equality follows from $\overline{\left(\frac{u}{v}\right)} = \frac{\bar{u}}{\bar{v}}$, which can be easily shown by Corollary 1.62. Then the proof is completed by Lemma 4.53. $\square$

Corollary 4.55. If $\mathcal{D} \subseteq \mathbb{C}$ is a domain with $\mathcal{D} \cap \mathbb{R} \neq \emptyset$ and if $f, g : \mathcal{D} \rightarrow \mathbb{C}$ are analytic functions with $f|_{\mathcal{D} \cap \mathbb{R}} = g|_{\mathcal{D} \cap \mathbb{R}}$, then we have $f(z) = g(z)$ for all $z \in \mathcal{D}$.

Proof. This follows directly from Theorem 4.31. $\square$

Exercise 4.56. Show that the real part and the imaginary part of an analytic function is real analytic in the sense of Definition B.102.

Lemma 4.57. Let $\mathcal{M} \subset \mathbb{R}$ be a non-degenerate interval. A function $f : \mathcal{M} \rightarrow \mathbb{R}$ has an analytic continuation to a domain $\mathcal{D} \subset \mathbb{C}$ if and only if f is real analytic in the sense of Definition B.102.

Proof. The necessity follows directly from Theorem 4.1 and Definitions 4.51 and B.102.

As for the sufficiency, we choose for any $a \in \mathcal{M}$ a number $\epsilon(a) > 0$ such that f is represented by its Taylor series in the interval $I_a := (a - \epsilon, a + \epsilon) \cap \mathcal{M}$. It follows from Abel's Theorem 1.77 that this Taylor series converges for all points within the open disk $\mathcal{U}_{\epsilon(a)}(a)$. By Theorem 4.9 (A) $\Leftrightarrow$ (F), we know that f is also analytic in $\mathcal{U}_{\epsilon(a)}(a)$.

For another point $b \in \mathcal{M} \setminus \{a\}$, arguments similar to the previous paragraph imply that f is analytic in $\mathcal{U}_{\epsilon(b)}(b)$. We can choose b to be sufficiently close to a such that $\mathcal{U}_{\epsilon(a)}(a) \cap \mathcal{U}_{\epsilon(b)}(b) \neq \emptyset$. Then it follows from the identity theorem 4.31 that f can be analytically extended from $\mathcal{U}_{\epsilon(a)}(a)$ to $\mathcal{U}_{\epsilon(a)}(a) \cup \mathcal{U}_{\epsilon(b)}(b)$.

The proof is completed by define $\mathcal{D} := \bigcup_{z \in \mathcal{M}_p} \mathcal{U}_{\epsilon(z)}(z)$ where $\mathcal{M}_p$ is a finite set of points in $\mathcal{M}$ whose open balls constructed in the previous paragraph cover $\mathcal{M}$. $\square$

Corollary 4.58. Let $\mathcal{D} \subset \mathbb{C}$ be a domain, $\mathcal{M} \subset \mathcal{D}$ a subset with at least one limit point in $\mathcal{D}$, and $f : \mathcal{M} \rightarrow \mathbb{C}$ an analytic function. If there exists a function $F : \mathcal{D} \rightarrow \mathbb{C}$ as the analytic continuation of f , then F is unique.

Proof. This follows from the identity theorem 4.31. $\square$

4.5 Singularities

Definition 4.59. Let $\mathcal{D}$ be a domain and $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function. A point $a \notin \mathcal{D}$ is said to be an (*isolated*) *singularity* of the function f iff

$$\exists r > 0 \text{ s.t. } \dot{\mathcal{U}}_r(a) := \{z \in \mathbb{C} : |z - a| \in (0, r)\} \subseteq \mathcal{D}.$$

Definition 4.60. Let $\mathcal{D} \subset \mathbb{C}$ be an open set. A singularity $a \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ is called *removable* iff f can be analytically extended to $\mathcal{D} \cup \{a\}$, i.e. iff there exists an analytic continuation $F : \mathcal{D} \cup \{a\} \rightarrow \mathbb{C}$ with $F|_{\mathcal{D}} = f$.

Theorem 4.61 (Riemann removability condition). Let $\mathcal{D} \subset \mathbb{C}$ be an open set. A singularity $a \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ is removable iff there exists a punctured disk $\dot{\mathcal{U}}_r(a) \subseteq \mathcal{D}$ where f is bounded.

Proof. The necessity is obvious, and we only prove the sufficiency. WLOG, we assume $a = 0$. Construct a function $h : \mathcal{U}_r(0) \rightarrow \mathbb{C}$,

$$h(z) = \begin{cases} z^2 f(z), & z \neq 0; \\ 0, & z = 0. \end{cases}$$

h is analytic at 0 because

$$\lim_{z \rightarrow 0} \frac{h(z) - h(0)}{z - 0} = \lim_{z \rightarrow 0} z f(z) = 0.$$

Thus h is analytic in $\mathcal{U}_r(0)$ and Theorem 4.1 yields

$$h(z) = a_0 + a_1 z + a_2 z^2 + \cdots$$

Then $h(0) = h'(0) = 0$ implies $a_0 = a_1 = 0$. For all $z \neq 0$, the power series $a_2 + a_3 z + a_4 z^2 + \cdots$ defines an analytic function F that is also analytic at $z = 0$. The proof is then completed by Definition 4.60. $\square$

Example 4.62. The singularity $z = 0$ of $f(z) = \frac{\sin z}{z}$ is removable: $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$ implies that the condition of Theorem 4.61 holds.

Exercise 4.63. Using the concept of removable singularities and Cauchy integral formula, we can prove an important identity of the following Dirichlet integral

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}. \quad (4.18)$$

Make sure you check your answer against that on Page 140 of the textbook by Freitag and Busam.

Example 4.64. The singularity $z = 0$ is not removable for $f(z) = \frac{1}{z}$ and $g(z) = \sin \frac{1}{z}$.

Definition 4.65. Let $\mathcal{D} \subset \mathbb{C}$ be an open set. A singularity $a \notin \mathcal{D}$ of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ is called *non-essential* iff there exists an integer $m \in \mathbb{Z}$ such that the function

$$g(z) = (z - a)^m f(z) \quad (4.19)$$

has a removable singularity at a .

Lemma 4.66. Let $\mathcal{D} \subset \mathbb{C}$ be an open set and $a \notin \mathcal{D}$ a non-essential singularity of an analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$. If f does not vanish identically in some neighborhood of a , then there exists a minimal integer $k \in \mathbb{Z}$ such that the function

$$h(z) = (z - a)^k f(z) \quad (4.20)$$

has a removable singularity at a and $h(a) \neq 0$.

Proof. If a is a non-essential singularity of f , then by Definitions 4.65 and 4.60 the function $g(z)$ in (4.19) can be expanded as a power series at a . Furthermore, f does not vanish identically in some neighborhood of a , hence

$$\begin{aligned} \exists n \in \mathbb{N} \text{ s.t. } a_n \neq 0 \text{ and} \\ g(z) = a_n(z - a)^n + a_{n+1}(z - a)^{n+1} + \cdots \end{aligned} \quad (4.21)$$